ILONA BERGQUIST-MAHONEY

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SUMMARY

Research-adjacent software engineer — three years building numerical and data-intensive systems, with a background in applied mathematics. Prior work spans simulation tooling, scientific workflow orchestration, and the unglamorous plumbing that makes an efficient pipeline stay efficient once it meets real data. Interested in roles where correctness matters more than velocity; comfortable reading a paper and shipping the thing it describes.

EXPERIENCE

Software Engineer, Simulation Tools *Sept. 2023 – present*
Cairnwell Robotics Laboratory — Ostmark, MN

- Rewrote the contact-solver inner loop in C++17 with explicit SIMD intrinsics, giving a 3.4× speed-up on the standard benchmark suite and cutting an overnight batch from 9 h 20 m to 2 h 45 m.
- Built "Ferryman", the workflow scheduler that now runs every nightly regression — about 1,100 jobs — across a 48-node cluster; job failure rate fell from 7.2% to 0.9%.
- Authored the team's numerical-stability test harness after a floating-point regression shipped undetected; it has caught eleven such regressions since.
- Maintain the Python bindings (pybind11) used by the research staff, including the affiliate group at Ostmark University.

Data Engineer *June 2022 – Aug. 2023*
Halcyon Data Laboratory — Remote

- Designed the ingestion workflow for a 12 TB instrument archive; the incremental loader reduced the full-refresh window from 31 hours to roughly 70 minutes.
- Replaced a hand-rolled scheduler with Airflow and wrote the migration guide — 60-odd DAGs moved with no data loss and a single half-day of downtime.
- Cut cloud storage cost by 42% (≈ $18,400/yr) through Parquet conversion, column pruning, and a tiered retention policy the team had been putting off since 2020.

Research Assistant, Numerical Methods Group *Jan. 2021 – May 2022*
Ostmark University, Dept. of Applied Mathematics

- Implemented and validated an adaptive quadrature scheme in Julia; results appear in Bergquist-Mahoney & Sandoval, "Adaptive Quadrature for Stiff Kernels", *Journal of Computational Practice* 14(2), 2022, pp. 211–229.
- Ran the group's reproducibility audit — every figure in three papers regenerated from source in a single make target.

EDUCATION

M.S. Applied Mathematics *2020 – 2022*
Ostmark University — Ostmark, MN
Thesis: "Stiffness-Aware Step Control for Coupled Systems". Advisor: Prof. R. Sandoval.

B.A. Mathematics & Computer Science *2016 – 2020*
Havering College — Havering, WI
Magna cum laude · Putnam competition team, 2018–2019 · GPA 3.83/4.00

TECHNICAL SKILLS

Languages — C++17/20, Python, Julia, Rust (intermediate), SQL, Fortran (legacy maintenance), LaTeX

Scientific — NumPy, SciPy, Eigen, pybind11, OpenMP, MPI, CUDA (basic), BLAS/LAPACK

Data & workflow — Apache Airflow, Parquet, DuckDB, PostgreSQL, dbt, Snakemake

Tooling — CMake, Git, Docker, Slurm, GitHub Actions, perf, Valgrind, Google Benchmark

SELECTED PROJECTS

quadfellow — adaptive quadrature library *2022 – present*
github.com/ibergquist/quadfellow
A header-only C++ library implementing the stiffness-aware scheme from my thesis, with a thin Python wrapper. Used by two university groups and one affiliate lab; 430 stars. The documentation — written in a literate style — doubles as the reproducibility appendix for the 2022 paper.

plainmake — a reproducible-figures toolchain *2021 – 2023*

bergquist.example.dev/plainmake

A small tool that turns a paper's figure scripts into a single, cached, efficient build graph, so "make figures" regenerates every plot from raw data in one step — or fails loudly.